

BCS Preliminary Mathematics

Mathematical Reasoning: Secondary-Level Mastery

A formula-first, pattern-first and practice-oriented
guide

Prepared for the Fresh Graduate IT preparation repository.
Original synthesis based on the supplied NCTB Class 8 Mathematics,
Class 9-10 Mathematics, and Class 9-10 Higher Mathematics
textbooks, cross-checked against the BPSC mathematical-reasoning
syllabus.

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How to use this book

This is not a transcription of the supplied textbooks. It is an original BCS-focused synthesis of the concepts, formulas, problem patterns and methods needed to solve secondary-level mathematical-reasoning questions quickly and reliably. The supplied books provide the mathematical foundation; this guide reorganizes that foundation around examination use.

The 2026 BPSC preliminary syllabus allocates 15 marks to Mathematical Reasoning and groups the content into five broad areas: number and commercial arithmetic; algebra; indices, logarithms, sequences and series; geometry and mensuration; and sets, permutations/combinations, statistics and probability. [1] The BPSC written syllabus likewise describes mathematical reasoning as application of mathematics acquired up to secondary level and lists arithmetic, algebra, geometry, trigonometry, Cartesian geometry, sets, counting and elementary probability. [2]

The supplied Class 9–10 Higher Mathematics textbook contains chapters on Set and Function, Algebraic Expressions, Geometry, Geometric Construction, Equations, Inequalities, Infinite Series, Trigonometry, Exponential and Logarithmic Functions, Binomial Expansion, Coordinate Geometry, Plane Vectors, Solid Geometry and Probability. The Class 9–10 Mathematics textbook contains Real Numbers, Set and Function, Algebraic Expressions, Indices and Logarithms, One-Variable Equations, Lines/Angles/Triangles, Practical Geometry, Circles, Trigonometric Ratios, Distance and Height, Ratio and Proportion, Simultaneous Linear Equations, Finite Series, Similarity/Symmetry, Area Theorems, Mensuration and Statistics. The Class 8 textbook adds patterns, profit, measurement, algebraic identities and applications, algebraic fractions, simultaneous equations, sets, quadrilaterals, Pythagoras, circles and data.

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Chapter 1

The BCS Mathematics Map

1.1 What the official syllabus actually asks

For the preliminary examination, the official BPSC syllabus lists five three-mark areas under Mathematical Reasoning: real numbers, LCM/GCD, percentage, interest, ratio/proportion and profit/loss; algebraic identities, factorization, equations, inequalities and simultaneous equations; indices, logarithms and arithmetic/geometric sequences and series; geometry, Pythagoras, circles and mensuration; and sets, permutations/combinations, statistics and probability. [1]

Table 1.1: BCS preliminary syllabus mapped to the supplied textbooks.

BCS area	Main textbook foundations	High-value subskills
Arithmetic	Class 8; Class 9–10 Math	divisibility, fractions, percentage, ratio, average, interest, profit/loss, work/time
Algebra	Class 8; Class 9–10 Math; Higher Math	identities, factorization, equations, inequalities, polynomials
Indices, logs, series	Class 9–10 Math; Higher Math	exponent laws, logarithms, AP, GP, finite and infinite series
Geometry and mensuration	Class 8; Class 9–10 Math; Higher Math	triangles, circles, Pythagoras, area, volume, coordinate geometry, trigonometry
Sets, counting, probability, statistics	Class 8; Higher Math; Class 9–10 Math	Venn diagrams, counting, permutations, combinations, probability, central tendency

1.2 A practical solving hierarchy

When a question looks unfamiliar, classify it before calculating:

1. Identify the quantities and their units.
2. Translate the wording into an equation, ratio, diagram or counting model.
3. Look for a standard identity or invariant before expanding anything.
4. Reduce fractions and ratios early.
5. Estimate the answer range before committing to an option.
6. Use the shortest valid method, but verify with substitution or an independent check.

Exam tip. In MCQ mathematics, an exact method is not automatically the fastest method. Factorization, ratio cancellation,

option substitution, symmetry and bounds can eliminate most arithmetic.

Chapter 2

Number Systems, Divisibility and Real Numbers

2.1 Core number sets

The standard hierarchy is

$$\mathbb{N} \subseteq \mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R},$$

where $\mathbb{N}$ denotes natural numbers, $\mathbb{Z}$ integers, $\mathbb{Q}$ rational numbers and $\mathbb{R}$ real numbers. Irrational numbers are real but not rational.

2.2 Prime factorization

Every integer greater than 1 can be decomposed into prime factors. This is the basis of efficient HCF/GCD and LCM calculations.

If $a = \prod p_i^{\alpha_i}$ and $b = \prod p_i^{\beta_i}$, then

$$\gcd(a, b) = \prod p_i^{\min(\alpha_i, \beta_i)}, \quad \text{lcm}(a, b) = \prod p_i^{\max(\alpha_i, \beta_i)}.$$

For positive integers,

$$ab = \gcd(a, b) \text{lcm}(a, b).$$

2.3 Divisibility tests

- 2: last digit is even.
- 3: digit sum is divisible by 3.
- 4: last two digits are divisible by 4.
- 5: last digit is 0 or 5.
- 6: divisible by both 2 and 3.
- 8: last three digits are divisible by 8.
- 9: digit sum is divisible by 9.
- 10: last digit is 0.
- 11: alternating digit sum is divisible by 11.

Worked example. Find the smallest positive integer divisible by 12, 18 and 30. Prime factors are $12 = 2^2 \cdot 3$, $18 = 2 \cdot 3^2$, $30 = 2 \cdot 3 \cdot 5$. Therefore

$$\text{lcm} = 2^2 \cdot 3^2 \cdot 5 = 180.$$

Answer: 180.

2.4 Remainders

If $a = bq + r$, then r is the remainder on division by b , with $0 \leq r < b$. For modular arithmetic,

$$a \equiv r \pmod{m}.$$

Useful consequences include

$$(a + b) \bmod m = ((a \bmod m) + (b \bmod m)) \bmod m,$$

and similarly for multiplication.

Exam tip. For a large power, reduce the base modulo the divisor and look for a cycle.

Chapter 3

Fractions, Ratio, Proportion and Unitary Method

3.1 Fractions

For nonzero denominators,

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}, \quad \frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}.$$

Cancel common factors before multiplying.

3.2 Ratio

If $a : b = m : n$, then $a = km$ and $b = kn$ for some common factor k .

If a total T is divided in the ratio $m : n$, the shares are

$$\frac{m}{m+n}T, \quad \frac{n}{m+n}T.$$

For three parts $a : b : c$, divide by $a + b + c$.

3.3 Direct and inverse proportion

Direct proportion: $y \propto x \Rightarrow y = kx$. Inverse proportion: $y \propto 1/x \Rightarrow xy = k$.

Worked example. A sum is divided among A, B and C in the ratio 2 : 3 : 5. If the total is 2400, find C's share.

$$C = \frac{5}{2+3+5}(2400) = 1200.$$

Answer: 1200.

Chapter 4

Percentage, Average, Profit, Loss and Interest

4.1 Percentage

$$x\% \text{ of } y = \frac{x}{100}y.$$

An increase by $r\%$ multiplies a quantity by $1 + r/100$; a decrease by $r\%$ multiplies it by $1 - r/100$.

Two successive percentage changes $a\%$ and $b\%$ have net multiplier

$$\left(1 + \frac{a}{100}\right) \left(1 + \frac{b}{100}\right).$$

Thus the net percentage change is

$$a + b + \frac{ab}{100}\%$$

when both are increases; signs must be retained for decreases.

4.2 Average

$$\text{Average} = \frac{\text{sum of observations}}{\text{number of observations}}.$$

If the average of n observations is A , their sum is nA . If one value changes by d , the average changes by d/n .

4.3 Profit and loss

$$\text{Profit} = SP - CP, \quad \text{Loss} = CP - SP.$$

$$\text{Profit}\% = \frac{SP - CP}{CP} \times 100, \quad \text{Loss}\% = \frac{CP - SP}{CP} \times 100.$$

If marked price is M and discount is $d\%$,

$$SP = M \left(1 - \frac{d}{100} \right).$$

4.4 Simple and compound interest

For principal P , annual rate $r\%$, time n years,

$$SI = \frac{Prn}{100}, \quad A = P + SI.$$

For annual compounding,

$$A = P \left(1 + \frac{r}{100} \right)^n, \quad CI = A - P.$$

For depreciation at rate r , replace the plus sign by a minus sign.

Worked example. A price is increased by 20% and then reduced by 20%. What is the net change? Multiplier = $1.2 \times 0.8 = 0.96$. Therefore the final value is 96% of the original. **Answer:** A 4% decrease.

Chapter 5

Work, Time, Speed and Mixture Models

5.1 Work rate method

If A completes a job in x days, A's one-day rate is $1/x$. If A and B take x and y days separately,

$$\text{combined rate} = \frac{1}{x} + \frac{1}{y} = \frac{x+y}{xy}.$$

For three workers, add the three rates.

5.2 Pipes and cisterns

Treat filling as positive rate and emptying as negative rate. If a pipe fills in x hours and a drain empties in y hours,

$$r = \frac{1}{x} - \frac{1}{y}.$$

5.3 Speed, distance and time

$$v = \frac{d}{t}, \quad d = vt, \quad t = \frac{d}{v}.$$

For equal distances at speeds u and v , the average speed is the harmonic mean:

$$v_{avg} = \frac{2uv}{u+v}.$$

For equal time intervals, the average speed is the arithmetic mean.

Conversion:

$$1 \text{ m/s} = \frac{18}{5} \text{ km/h}, \quad 1 \text{ km/h} = \frac{5}{18} \text{ m/s}.$$

5.4 Mixture and weighted average

If quantities x_i have values/rates r_i , their weighted average is

$$\bar{r} = \frac{\sum x_i r_i}{\sum x_i}.$$

For two prices p_1, p_2 mixed in quantities q_1, q_2 , the unit price is

$$\frac{p_1 q_1 + p_2 q_2}{q_1 + q_2}.$$

Chapter 6

Algebraic Identities and Factorization

The high-frequency identities are

$$\begin{aligned}(a+b)^2 &= a^2 + 2ab + b^2, \\(a-b)^2 &= a^2 - 2ab + b^2, \\a^2 - b^2 &= (a-b)(a+b), \\(a+b)^3 &= a^3 + 3a^2b + 3ab^2 + b^3, \\(a-b)^3 &= a^3 - 3a^2b + 3ab^2 - b^3, \\a^3 + b^3 &= (a+b)(a^2 - ab + b^2), \\a^3 - b^3 &= (a-b)(a^2 + ab + b^2).\end{aligned}$$

A particularly useful identity is

$$a^2 + b^2 = (a+b)^2 - 2ab.$$

Thus knowing $a+b$ and ab is often enough to obtain $a^2 + b^2$, $a^3 + b^3$, or the roots of a quadratic.

6.1 Quadratic equations

For $ax^2 + bx + c = 0$,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

The discriminant is $D = b^2 - 4ac$.

$$x_1 + x_2 = -\frac{b}{a}, \quad x_1 x_2 = \frac{c}{a}.$$

Worked example. If $x + y = 7$ and $xy = 10$, find $x^2 + y^2$.

$$x^2 + y^2 = (x + y)^2 - 2xy = 49 - 20 = 29.$$

Answer: 29.

Chapter 7

Linear Equations, Simultaneous Equations and Inequalities

7.1 Linear equations

Preserve equality when manipulating an equation. Clear denominators only after identifying excluded values.

7.2 Two linear equations

For

$$a_1x + b_1y = c_1, \quad a_2x + b_2y = c_2,$$

elimination is usually fastest. Cramer's rule gives

$$D = a_1b_2 - a_2b_1,$$
$$x = \frac{c_1b_2 - c_2b_1}{D}, \quad y = \frac{a_1c_2 - a_2c_1}{D},$$

when $D \neq 0$.

7.3 Inequalities

Adding or subtracting the same quantity preserves inequality direction. Multiplying or dividing by a negative quantity reverses

it. For quadratic inequalities, find the roots, partition the number line into intervals, and test the sign of each interval.

Exam tip. Never divide an inequality by an expression whose sign has not been established.

Chapter 8

Indices and Logarithms

8.1 Index laws

For appropriate nonzero bases,

$$\begin{aligned}a^m a^n &= a^{m+n}, & \frac{a^m}{a^n} &= a^{m-n}, \\(a^m)^n &= a^{mn}, & (ab)^n &= a^n b^n, \\a^0 &= 1, & a^{-n} &= \frac{1}{a^n}.\end{aligned}$$

Fractional powers satisfy $a^{m/n} = \sqrt[n]{a^m}$ where defined.

8.2 Logarithms

$$\log_a x = y \iff a^y = x, \quad a > 0, a \neq 1, x > 0.$$

Rules:

$$\begin{aligned}\log_a(xy) &= \log_a x + \log_a y, \\ \log_a \left(\frac{x}{y} \right) &= \log_a x - \log_a y, \\ \log_a(x^k) &= k \log_a x, \\ \log_a b &= \frac{\log_c b}{\log_c a}.\end{aligned}$$

Important special values: $\log_a 1 = 0$ and $\log_a a = 1$.

Worked example. Solve $2^{x+1} = 16$. Since $16 = 2^4$, $x+1 = 4$, hence $x = 3$. **Answer:** 3.

Chapter 9

Arithmetic and Geometric Sequences and Series

9.1 Arithmetic progression

For first term a , common difference d , n th term:

$$a_n = a + (n - 1)d.$$

Sum of first n terms:

$$S_n = \frac{n}{2}[2a + (n - 1)d] = \frac{n}{2}(a + l),$$

where l is the last term.

9.2 Geometric progression

For first term a , common ratio r ,

$$a_n = ar^{n-1}.$$

For $r \neq 1$,

$$S_n = a \frac{r^n - 1}{r - 1} = a \frac{1 - r^n}{1 - r}.$$

If $|r| < 1$, the infinite sum is

$$S_\infty = \frac{a}{1 - r}.$$

9.3 Useful sequence observations

- Constant first difference suggests an AP.
- Constant ratio suggests a GP.
- Constant second difference suggests a quadratic pattern.
- A sequence where each term depends on the previous two terms often signals a recurrence such as Fibonacci.

Chapter 10

Set Theory, Venn Diagrams and Counting

10.1 Sets

For sets A, B , union is $A \cup B$, intersection is $A \cap B$, difference is $A \setminus B$, and complement is A^c relative to the universal set.

For two finite sets,

$$n(A \cup B) = n(A) + n(B) - n(A \cap B).$$

For three sets,

$$\begin{aligned} n(A \cup B \cup C) &= n(A) + n(B) + n(C) \\ &\quad - n(A \cap B) - n(B \cap C) - n(C \cap A) \\ &\quad + n(A \cap B \cap C). \end{aligned}$$

10.2 Counting principles

If a task has m choices followed by n choices, there are mn ordered outcomes.

10.3 Permutations and combinations

$${}_nP_r = \frac{n!}{(n-r)!}, \quad {}_nC_r = \frac{n!}{r!(n-r)!}.$$

Relations:

$${}^nC_r = {}^nC_{n-r}, \quad {}^nP_r = r! {}^nC_r.$$

Use permutations when order matters; combinations when it does not.

Worked example. How many different three-person ordered committees can be selected from eight people?

$${}^8P_3 = 8 \cdot 7 \cdot 6 = 336.$$

Answer: 336.

Chapter 11

Probability

For equally likely outcomes,

$$P(E) = \frac{n(E)}{n(S)}.$$

Always satisfy $0 \leq P(E) \leq 1$.

Complement:

$$P(E^c) = 1 - P(E).$$

For mutually exclusive events,

$$P(A \cup B) = P(A) + P(B).$$

In general,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

Conditional probability:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}.$$

If A and B are independent,

$$P(A \cap B) = P(A)P(B).$$

Exam tip. In card, dice and coin questions, define the sample space before counting favorable cases. For "at least one", the complement is often shorter.

Chapter 12

Geometry: Lines, Angles and Triangles

12.1 Angle facts

- Straight angle: 180° .
- Full angle: 360° .
- Vertically opposite angles are equal.
- Complementary angles sum to 90° .
- Supplementary angles sum to 180° .

For parallel lines cut by a transversal, corresponding and alternate interior angles are equal; co-interior angles are supplementary.

12.2 Triangles

$$A + B + C = 180^\circ.$$

Exterior angle equals the sum of the two opposite interior angles.
For a right triangle,

$$a^2 + b^2 = c^2.$$

The area is

$$\frac{1}{2}bh.$$

Heron's formula, useful when all three sides are known:

$$s = \frac{a + b + c}{2}, \quad K = \sqrt{s(s - a)(s - b)(s - c)}.$$

12.3 Similarity

If two triangles are similar, corresponding sides are proportional and corresponding angles equal. If the side scale factor is k , areas scale by k^2 .

Chapter 13

Quadrilaterals, Polygons and Circles

For an n -gon,

$$\text{sum of interior angles} = (n - 2)180^\circ.$$

A parallelogram has opposite sides and opposite angles equal; diagonals bisect each other. A rectangle has four right angles. A rhombus has four equal sides. A square has both rectangle and rhombus properties.

13.1 Circle essentials

Circumference:

$$C = 2\pi r.$$

Area:

$$A = \pi r^2.$$

Arc length for central angle θ degrees:

$$L = \frac{\theta}{360^\circ} 2\pi r.$$

Sector area:

$$A_s = \frac{\theta}{360^\circ} \pi r^2.$$

An angle subtended by a diameter is a right angle. Equal chords subtend equal angles at the center; the perpendicular from the center to a chord bisects the chord.

Chapter 14

Mensuration: Plane Figures and Solids

14.1 Plane figures

Table 14.1: Core plane-area formulas.

Figure	Area	Perimeter/circumference
Square, side a	a^2	$4a$
Rectangle $l \times w$	lw	$2(l + w)$
Triangle	$\frac{1}{2}bh$	sum of sides
Parallelogram	bh	$2(a + b)$
Circle	πr^2	$2\pi r$
Trapezium	$\frac{1}{2}(a + b)h$	sum of sides

14.2 Solids

$$V_{\text{cuboid}} = lwh,$$

$$V_{\text{cube}} = a^3,$$

$$V_{\text{cylinder}} = \pi r^2 h,$$

$$V_{\text{cone}} = \frac{1}{3}\pi r^2 h,$$

$$V_{\text{sphere}} = \frac{4}{3}\pi r^3.$$

Surface areas:

$$SA_{sphere} = 4\pi r^2, \quad SA_{cylinder, curved} = 2\pi rh.$$

Exam tip. Keep units consistent. A common trap is mixing square units, cubic units, metres and centimetres.

Chapter 15

Trigonometry and Height-Distance

For a right triangle with angle θ ,

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}, \quad \tan \theta = \frac{\text{opposite}}{\text{adjacent}}.$$

Identities:

$$\sin^2 \theta + \cos^2 \theta = 1, \quad 1 + \tan^2 \theta = \sec^2 \theta.$$

Standard values:

Table 15.1: Standard trigonometric values.

θ	0°	30°	45°	60°	90°
$\sin \theta$	0	$1/2$	$1/\sqrt{2}$	$\sqrt{3}/2$	1
$\cos \theta$	1	$\sqrt{3}/2$	$1/\sqrt{2}$	$1/2$	0
$\tan \theta$	0	$1/\sqrt{3}$	1	$\sqrt{3}$	undefined

15.1 Height and distance

Draw the right triangle first. If the observer is at horizontal distance d from the foot of an object and angle of elevation is θ , then

$$h = d \tan \theta.$$

If the line of sight is the hypotenuse L , then $h = L \sin \theta$ and $d = L \cos \theta$.

Chapter 16

Coordinate Geometry and Graphs

Distance between (x_1, y_1) and (x_2, y_2) :

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

Midpoint:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

Slope:

$$m = \frac{y_2 - y_1}{x_2 - x_1}.$$

Straight-line forms:

$$y = mx + c, \quad y - y_1 = m(x - x_1).$$

Parallel nonvertical lines have equal slopes; perpendicular lines satisfy $m_1 m_2 = -1$.

16.1 Graph strategy

For a linear equation, find the intercepts or two convenient points. For MCQs, checking a proposed point by substitution is often faster than drawing the graph.

Chapter 17

Statistics and Data

17.1 Central tendency

Mean:

$$\bar{x} = \frac{\sum x_i}{n}.$$

For frequency data,

$$\bar{x} = \frac{\sum f_i x_i}{\sum f_i}.$$

Median is the middle ordered observation; for an even number of observations it is the average of the two central observations. Mode is the most frequent observation.

17.2 Range and basic dispersion

Range = maximum – minimum.

When an observation is added or removed, update the total sum rather than recomputing every value.

Chapter 18

Algebraic and Numerical Shortcuts for MCQs

18.1 Option substitution

If the question asks for a value of x and gives four choices, testing the choices directly can be faster than solving a high-degree equation.

18.2 Back-solving

For word problems involving totals, prices, ages or rates, insert each candidate into the original condition. Reject candidates that violate even one condition.

18.3 Approximation and bounds

If the answer choices are widely separated, estimate. For positive a, b ,

$$2\sqrt{ab} \leq a + b$$

by AM-GM. This can rapidly eliminate impossible options.

18.4 Difference of squares

If a calculation resembles $n^2 - k^2$, use

$$n^2 - k^2 = (n - k)(n + k).$$

For example,

$$1001^2 - 999^2 = (2)(2000) = 4000.$$

18.5 Near-100 multiplication

For numbers near 100,

$$(100 + a)(100 + b) = 10000 + 100(a + b) + ab.$$

This reduces mental multiplication.

Chapter 19

BCS Question Archetypes

The historical BCS solution collection supplied by the user is a useful practice index for 10th–40th BCS preliminary mathematics. Public descriptions of that collection identify it as a sequence of question-solution resources for the 10th through 40th examinations. [3]

The purpose of this chapter is not to reproduce those copyrighted question sets. Instead, it extracts the recurring *problem forms* that should be mastered.

Table 19.1: High-frequency question archetypes to drill.

Archetype	Recognition rule
LCM/GCD	prime factorization, divisibility, repeated cycles
Percentage chain	successive increase/decrease, discount, population/value change
Ratio sharing	total divided into integer parts
Average replacement	one observation replaced or added
Profit or loss	CP, SP, MP and discount relations
Interest	SI/CI, rate/time/principal
Work rate	independent completion times, pipes and drains
Speed, time and distance	relative motion, trains, equal-distance average speed
Identity or factorization	hidden $(a \pm b)^2$, cubes, difference of squares
Quadratic roots	sum/product of roots, discriminant, direct solution
Indices and logs	common-base transformation, log laws
AP or GP	nth term, sum, missing term, ratio/difference
Geometry theorem	angle chase, similarity, Pythagoras, circle facts
Mensuration	area/volume plus unit conversion
Trigonometry	standard-angle values and height/distance
Set counting	union/intersection/complement and Venn diagrams
Permutation or combination	order matters or does not
Probability	complement, mutually exclusive/independent events
Statistics	mean, median, mode, frequency table

Chapter 20

Speed Sheet: Formulas to Memorize

- $ab = \gcd(a, b) \operatorname{lcm}(a, b)$
- $\text{percentage} = \frac{\text{part}}{\text{whole}} 100$
- $\bar{x} = \frac{\sum x}{n}$
- $SP = CP(1 \pm r/100)$
- $SI = Prn/100$
- $A_{CI} = P(1 + r/100)^n$
- $\text{rate} = 1/\text{time}$
- $v = d/t$
- $u + v$ for combined direct rates
- $(a + b)^2 = a^2 + 2ab + b^2$
- $a^2 - b^2 = (a - b)(a + b)$
- $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$
- $x_1 + x_2 = -b/a$
- $x_1 x_2 = c/a$
- $a^m a^n = a^{m+n}$
- $\log_a xy = \log_a x + \log_a y$
- $a_n = a + (n - 1)d$
- $S_n = n[2a + (n - 1)d]/2$
- $a_n = ar^{n-1}$
- $S_\infty = a/(1 - r)$
- $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
- ${}^n P_r = n!/(n - r)!$
- ${}^n C_r = n!/[r!(n - r)!]$
- $P(E^c) = 1 - P(E)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $a^2 + b^2 = c^2$
- $A_{\text{triangle}} = bh/2$
- $C = 2\pi r$
- $A_{\text{circle}} = \pi r^2$
- $d = \sqrt{(\Delta x)^2 + (\Delta y)^2}$
- $m = \Delta y / \Delta x$
- $\sin^2 \theta + \cos^2 \theta = 1$
- $\tan \theta = \sin \theta / \cos \theta$
- $V_{\text{cylinder}} = \pi r^2 h$
- $V_{\text{sphere}} = 4\pi r^3 / 3$

Chapter 21

Original Practice Set I: Fundamentals

Answer each without looking at the answer key. These are original practice questions designed around the syllabus and textbook foundations.

- Q1.** Find the LCM of 24, 36 and 54.
- Q2.** A number is increased by 15% and then decreased by 15%. Find the net percentage change.
- Q3.** The ratio of two numbers is 3 : 5 and their sum is 64. Find the larger number.
- Q4.** The average of five numbers is 18. If one number, 10, is replaced by 25, find the new average.
- Q5.** An article costing 800 is sold at a 12.5% profit. Find its selling price.
- Q6.** A principal of 5000 earns simple interest at 8% per year for 3 years. Find the interest.
- Q7.** A can complete a job in 12 days and B in 18 days. How long do they take together?
- Q8.** If $x + y = 9$ and $xy = 14$, find $x^2 + y^2$.
- Q9.** Solve $3x - 7 = 20$.
- Q10.** Solve $x^2 - 7x + 12 = 0$.

- Q11.** Simplify $2^3 \cdot 2^5/2^4$.
- Q12.** Solve $\log_2 x = 5$.
- Q13.** Find the 20th term of the AP 7, 10, 13, ...
- Q14.** Find the sum of the first 10 positive integers.
- Q15.** Find the number of diagonals of a hexagon.
- Q16.** The sides of a right triangle are 6 and 8 for the perpendicular sides. Find the hypotenuse.
- Q17.** Find the area of a circle of radius 7 using $\pi = 22/7$.
- Q18.** If $\sin \theta = 3/5$ for an acute angle, find $\cos \theta$.
- Q19.** Find the distance between $(1, 2)$ and $(4, 6)$.
- Q20.** A bag contains 3 red and 2 blue balls. One ball is selected at random. Find the probability of blue.

Chapter 22

Original Practice Set II: Mixed BCS Reasoning

- Q1.** The HCF of two positive integers is 12 and their LCM is 180. If one integer is 36, find the other.
- Q2.** A population rises by 10% and then by another 20%. Find the total percentage increase.
- Q3.** If 40% of a number is 72, find the number.
- Q4.** A shopkeeper marks an article 25% above cost and gives a 10% discount. Find the profit percentage.
- Q5.** A sum becomes 12100 in two years at 10% compound interest annually. Find the principal.
- Q6.** A train travels 180 km in 3 hours. Find its speed in m/s.
- Q7.** A works twice as fast as B. Together they finish a job in 12 days. How many days would A alone need?
- Q8.** If $a + b = 10$ and $a - b = 4$, find ab .
- Q9.** If the roots of $x^2 - 9x + k = 0$ have product 14, find k .
- Q10.** Solve $2^x = 32$.
- Q11.** Find the sum of the first 8 terms of the GP 3, 6, 12, ...
- Q12.** If A and B are disjoint and $P(A) = 0.3$, $P(B) = 0.2$, find $P(A \cup B)$.
- Q13.** From 7 people, how many unordered pairs can be selected?

- Q14.** From 7 people, how many ordered pairs can be selected without repetition?
- Q15.** A triangle has angles $2x, 3x, 4x$. Find the largest angle.
- Q16.** The radius of a circle is doubled. By what factor does its area change?
- Q17.** A cylinder has radius 3 and height 5. Express its volume in terms of π .
- Q18.** Find the equation of the line with slope 2 passing through $(1, 3)$.
- Q19.** If $\tan \theta = 3/4$ for acute θ , find $\sec \theta$.
- Q20.** The mean of 4 observations is 15. A fifth observation is 30. Find the new mean.

Chapter 23

Answer Key and Compact Solutions

23.1 Practice Set I

Q1. $2^3 \cdot 3^3 = 216$.

Q2. Multiplier $1.15 \times 0.85 = 0.9775$, so 2.25% decrease.

Q3. Larger number $= 64(5/8) = 40$.

Q4. Original sum $= 90$; new sum $= 105$; average $= 21$.

Q5. $800(1.125) = 900$.

Q6. $5000(0.08)(3) = 1200$.

Q7. Rate $= 1/12 + 1/18 = 5/36$, so time $= 36/5$ days.

Q8. $81 - 28 = 53$.

Q9. $x = 9$.

Q10. $(x - 3)(x - 4) = 0$, so $x = 3, 4$.

Q11. $2^4 = 16$.

Q12. $x = 32$.

Q13. $7 + 19(3) = 64$.

Q14. $10(11)/2 = 55$.

Q15. $6(6 - 3)/2 = 9$.

Q16. $\sqrt{36 + 64} = 10$.

Q17. $22/7 \cdot 49 = 154$.

Q18. $4/5$.

Q19. $\sqrt{3^2 + 4^2} = 5$.

Q20. $2/5$.

23.2 Practice Set II

Q1. $36x = 12 \cdot 180$, so $x = 60$.

Q2. $1.10 \times 1.20 = 1.32$, so 32% increase.

Q3. $72/0.4 = 180$.

Q4. Net multiplier $1.25 \times 0.90 = 1.125$, so 12.5% profit.

Q5. $P(1.1)^2 = 12100$, so $P = 10000$.

Q6. $180/3 = 60$ km/h $= 60(5/18) = 50/3$ m/s.

Q7. Let B's rate be r ; A's is $2r$. Then $3r = 1/12$, so A takes 18 days.

Q8. $a = (10 + 4)/2 = 7$, $b = 3$, so $ab = 21$.

Q9. Product of roots is k , so $k = 14$.

Q10. $x = 5$.

Q11. $3(2^8 - 1)/(2 - 1) = 765$.

Q12. $0.3 + 0.2 = 0.5$.

Q13. ${}^7C_2 = 21$.

Q14. ${}^7P_2 = 42$.

Q15. $9x = 180^\circ$, so largest angle $= 80^\circ$.

Q16. Area factor $= 2^2 = 4$.

Q17. $\pi(3^2)(5) = 45\pi$.

Q18. $y - 3 = 2(x - 1)$, so $y = 2x + 1$.

Q19. $\sec^2 \theta = 1 + 9/16 = 25/16$, hence $\sec \theta = 5/4$.

Q20. Original sum 60; new sum 90; mean 18.

Chapter 24

Final Revision Protocol

24.1 Seven-pass revision

1. Pass 1: memorize arithmetic identities, percentage multipliers and ratio rules.
2. Pass 2: drill algebraic identities, factorization and equations.
3. Pass 3: drill indices, logarithms, AP or GP and series.
4. Pass 4: draw geometry figures and reproduce core theorems from memory.
5. Pass 5: memorize mensuration and standard trigonometric values.
6. Pass 6: drill sets, counting, probability and statistics.
7. Pass 7: solve mixed MCQs under time pressure and classify every error by topic.

24.2 Error log template

Table 24.1: Recommended mathematics error log.

Date	Topic	Error type	Correct rule/method
		concept / arithmetic / interpretation / time	
		concept / arithmetic / interpretation / time	
		concept / arithmetic / interpretation / time	

Source Notes and Scope

24.3 Supplied textbooks

This guide was prepared from the supplied 2026 NCTB editions of:

- Class 8 Mathematics;
- Class 9–10 Mathematics;
- Class 9–10 Higher Mathematics.

The source textbooks are the basis for the mathematical terminology, progression and topic coverage. The present document intentionally paraphrases and synthesizes rather than reproducing textbook passages.

24.4 BCS sources

The current BPSC preliminary syllabus was used to select and prioritize the topics in this guide. The historical 10th–40th BCS math-solution collection supplied by the user was used only as a practice-index reference; questions are not reproduced here.

Bibliography

- [1] Bangladesh Public Service Commission, *Syllabus for BCS Preliminary Test*, Mathematical Reasoning, 15 marks. Official BPSC source: https://bpsc.gov.bd/sites/default/files/files/bpsc.portal.gov.bd/psc_exam/93992bad_97b2_4d13_84eb_902cc36f2d5f/SYLLABUS%20FOR%20BCS%20PRILIMINARY%20TEST.pdf.
- [2] Bangladesh Public Service Commission, *Syllabus for BCS (Written) Examination*, Mathematical Reasoning. Official BPSC source: https://bpsc.gov.bd/sites/default/files/files/bpsc.portal.gov.bd/psc_exam/3c11cd64_6370_4e04_be9a_60f82b085597/SYLLABUS%20FOR%20BCS%20%28WRITTEN%29%20EXAMINATION.pdf.
- [3] Public descriptions of the supplied 10th–40th BCS mathematics question/solution collection, including the Scribd index and related solution index pages. These are treated as practice-navigation sources, not as authoritative BPSC documents.